MTH636 – Game Theory

Lecture 5: Auctions (Cont.) & Max-Min Strategies

1 Auctions (Continued)

Recall from the previous class regarding the **Second-Price Sealed-Bid Auction**:

$$u_i(v_i, b_{-i}) \geq u_i(b_i', b_{-i}) \quad \forall b_{-i} \in [0, \infty)^{n-1},\ \forall b_i' \in [0, \infty)$$

This states that bidding your true valuation v_i weakly dominates any other bid b_i'.

Example: 3-Player Auction Example

Suppose there are three players. For Player 1, their true valuation is $v_1 = 11$. Assume the other bids are $b_2 = 10$ and $b_3 = 8$.

- If Player 1 bids $b_1 = 11$ (truthful), they win and pay $\max(10, 8) = 10$. Their payoff is $11 - 10 = 1$.
- If they bid lower, say $b_1 = 9$, they lose to Player 2 (who bid 10) and get a payoff of 0.
- If they bid higher, say $b_1 = 15$, they still win and still pay 10. Payoff is still 1.

Thus, deviating from the true valuation either doesn't change the payoff or strictly hurts the player.

Key Insight: First-Price & Third-Price Auctions

- The above theorem **does not hold in a First-Price Auction**. In a first-price auction, if you bid your true valuation and win, your payoff is $v_i - v_i = 0$. You are always incentivized to bid lower than your valuation (shade your bid) to obtain a positive surplus.
- **Exercise:** Assume $n \geq 3$ and check if the theorem holds for a **Third-Price Auction** (where the winner pays the third-highest bid).

2 Max-Min Value and Max-Min Strategies (Continued)

A player plays a strategy s_i. The worst-case payoff they could get is:

$$\min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$$

For player i, by playing a specific strategy, they can ensure they get at least:

$$\max_{s_i \in S_i} \left(\min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) \right)$$

This value is called the **max-min value** for player i, often denoted $\underline{v}_i$.

And any strategy s_i^* that achieves this, i.e.,

$$s_i^* \in \arg\max_{s_i \in S_i} \left(\min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) \right)$$

is called a **max-min strategy** for player i. If a player plays a max-min strategy, he/she will get at least his max-min value.

2.1 Infinite Strategy Spaces

If the strategy spaces are infinite, we use *supremum* and *infimum* instead of max and min. The max-min value is defined as:

$$\sup_{s_i \in S_i} \inf_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$$

Example: Guessing Game / Average Game

Players submit numbers x_1, x_2, x_3. The target $T = T(x_1, x_2, x_3)$ is some function (e.g., the average). The payoff $u_i(x_1, x_2, x_3)$ is 0 if someone else is closer to the target, and $\frac{1}{|\text{winners}|}$ if there is a tie for being closest.
What is Player 1's max-min value?

- If $x_1 = 10$, the opponents can always choose x_2, x_3 such that Player 1 loses. So $\inf_{x_2,x_3} u_1(10, x_2, x_3) = 0$.
- If $x_1 = 0$, similarly $\inf_{x_2,x_3} u_1(0, x_2, x_3) = 0$.

The max-min value is 0. Therefore, **any** strategy is a max-min strategy, because no strategy can guarantee a payoff strictly greater than 0.

Note

For the **second-price auction**, the max-min value is 0. Bidding your valuation guarantees at least 0 (you never pay more than your valuation). Playing anything less than your valuation is also a max-min strategy (since it also prevents you from losing money, guaranteeing at least 0).

3 Max-Min Values and Dominated Strategies

Suppose Player i's max-min value is $\underline{v}_i$ in game G. Let G' be a new game obtained by removing a dominated strategy.

Theorem: Effect of Eliminating Strategies

1. If we remove a strategy t_i of **Player** i that is dominated by some other strategy, Player i's max-min value in G' is **still exactly** $\underline{v}_i$. (Removing a bad option for yourself doesn't hurt your guaranteed payoff).
2. If we remove a strategy t_j of **Player** j (an opponent) that is dominated, Player i's max-min value in G' **may increase** (or stay the same, but it will not decrease). (Removing options from your opponent restricts how much they can hurt you).

4 Introduction to Two-Player Zero-Sum Games

Definition: Two-Player Zero-Sum Game

A game $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ is called a two-player zero-sum game if:

1. $N = \{1, 2\}$
2. $u_1(s_1, s_2) + u_2(s_1, s_2) = 0 \quad \forall (s_1, s_2) \in S_1 \times S_2$

This means whatever Player 1 wins, Player 2 loses. $u_2(s_1, s_2) = -u_1(s_1, s_2)$. Because of this, we often only write the payoffs for Player 1 in the matrix.

$P_1 \setminus P_2$	$s_{2,1}$	$s_{2,2}$
$s_{1,1}$	$u_1(s_{1,1}, s_{2,1})$	$u_1(s_{1,1}, s_{2,2})$
$s_{1,2}$	$u_1(s_{1,2}, s_{2,1})$	$u_1(s_{1,2}, s_{2,2})$